



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT4220
Course title (English)	Partial Differential Equations
Course title (Chinese)	偏微分方程
Units	3
Language of Instruction	English
Description (English)	This course introduces the theory of partial differential equations (PDEs) that models phenomena in areas including physics, biology, chemistry and social science. Solution techniques such as the separation of variables, eigenfunction expansions, Green's functions, Fourier and Laplace transforms, weak solutions and Sobolev spaces for solving those equations will be studied in detail.
Description (Chinese)	本科目介绍抛物型、椭圆型和双曲型偏微分方程及方程组，这类方程用于描述物理、生物、化学、社会科学等领域的模型。本科目主要注重于介绍三类基本方程包括Laplace方程，热方程以及波方程；学习求解这几类偏微分方程的基本方法，例如：分离变量法、特征函数展开法、Green函数、Fourier和Laplace变换。

2. Prerequisites / Co-requisites

A. Prerequisites

MAT2006 Elementary Real Analysis
MAT2002 Ordinary Differential Equation

B. Co-requisites

3. Learning Outcomes

Knowledge



- a) Students will understand the classification of second order partial differential equations.
- b) Students will understand the basic techniques in solving parabolic equations.
- c) Students will understand the basic techniques in solving elliptic equations.
- d) Students will understand the basic techniques in solving hyperbolic equations.
- e) Students will understand the basic concepts of characteristics for hyperbolic type equations.
- f) Students will understand some basic concepts of Fourier transforms.

Skills

Generic:

- a) Students will be able to recognize some second-order partial differential equations of mathematical physics, such as diffusion equation, Laplace equation and wave equation.
- b) Students will be able to solve typical second-order partial differential equations by the method of separation of variables.
- c) Students will be able to solve problems of elliptic or parabolic equations by constructing fundamental solutions.
- d) Students will be able to solve the Cauchy problem of the hyperbolic equation by the method of characteristics.
- e) Students will be able to apply the Fourier transform to solve linear partial differential equations.

Valued/ Attitude:

- a) Students will be able to formulate real life problems by partial differential equations.
- b) Students will be able to understand the partial differential equations as a unified tool in modeling physical world problems.
- c) Students will be able to understand how mathematics applied to other sciences, such as: biological, physical, and social sciences.

4. Course syllabus

Classification of second-order partial differential equations; Heat conduction equation; Initial boundary value problems; Uniqueness; Method of separation of variables; Eigenvalue problems; Cauchy problem and fundamental solution; Laplace's equation; Boundary value problems; Properties of harmonic functions; Green's function; Wave equation; characteristics; D'Alembert formula; Fourier transform.

5. Assessment Scheme



Component/ method	% weight
Mid-term Exam	30
Assignments	20
Final Exam	50

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

- A course questionnaire answered by every student after finishing all the lectures



- Reflections from teachers of more advanced courses
- Feedback from questions in class, informal conversations in office, and e-mail correspondence with students outside classroom

8. Reading

Required Reading

1.2, United States, W. A. Strauss, Partial Differential Equations: An Introduction, Wiley;
2nd edition (2007)

9. Course components

Activity	Hours/week
Lectures	3
Readings & Assignments	5
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	Introduction: Definitions and Examples
00000 00002	First-Order Equations: Characteristics
00000 00003	Classification of Second-Order Equations with Constant Coefficients
00000 00004	Wave Equation
00000 00005	Diffusion equation
00000 00006	Separation of Variables, boundary value Problems
00000 00007	Fourier series
00000 00008	Fourier series and Mid-term exam
00000 00009	Laplace equation
00000 00010	Laplace equation



00000 00011	Green's identities and Green's functions
00000 00012	Fourier and Laplace transforms
00000 00013	Remarks regarding more PDEs and techniques.
00000 00014	Final Review

11. Any other information

Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course.

12. Version date

Version number	3
As of (date)	2024/08/12